Applied Soft Computing Journal 169 (2025) 112596

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Learning with noisy labels via Mamba and entropy KNN framework

Ningwei Wang [a,1], Weiqiang Jin [a,1], Shirou Jing [b], Haixia Bi [a,*], Guang Yang [c,d,e,f]

[a] School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi, 710049, China
[b] University of Chinese Academy of Science, No. 19(A) Yuquan Road, Beijing, 100048, China
[c] Bioengineering Department and Imperial-X, Imperial College London, London, W12 7SL, UK
[d] National Heart and Lung Institute, Imperial College London, London, SW7 2AZ, UK
[e] Cardiovascular Research Centre, Royal Brompton Hospital, London, SW3 6NP, UK
[f] School of Biomedical Engineering & Imaging Sciences, King's College London, London, WC2R 2LS, UK

ARTICLE INFO

Keywords:
Label rectification
Mamba
Label selection
Entropy-variance
KNN
Adaptpuzzle

ABSTRACT

Learning from corrupted data marginally degrades model performance. As deep learning proliferates, the need for large, accurately labeled datasets becomes crucial. Central to this challenge is the concept of learning with noisy labels (LNL). A crucial element in achieving successful learning via this approach is an initial warm-up phase that strengthens the discriminative power of the backbone. To address this issue, we introduce RelabelMamba, a novel model designed for noisy data that employs image sequences with position embeddings and leverages bidirectional state space models to compress visual representations. In contrast to traditional methods that rely on Convolutional Neural Networks (CNNs) with limited local receptive fields or Vision Transformers (ViT) which provide global receptive fields at the expense of quadratic complexity, our model achieves global receptive fields coverage with linear complexity. Additionally, we introduce two innovative components: Augment Entropy-variance K-Nearest Neighbors Relabel (AEKR), a label correction strategy that uses feature similarity and entropy-variance uncertainty to precisely correct mislabeled samples by evaluating their proximity in feature space, assigning normalized weights to nearest neighbors, and utilizing a score matrix with a predefined threshold to identify clean samples. This method also augments the data categories with fewer corrected samples, bringing their counts up to the average. Adaptpuzzle, enhanced by a refined mixup strategy, adjusts the mixing coefficient based on the variance of the data to optimize the fusion of sample pairs and their labels. Our results have either surpassed or closely matched the state-of-the-art on the CIFAR-10/100, Clothing1M, and ANIMAL-10N datasets, demonstrating the efficacy of RelabelMamba in LNL.

1. Introduction

Deep learning has garnered immense popularity in artificial intelligence, particularly in computer vision, with applications spanning image classification [1,2], object detection [3,4], image segmentation [5–7], and visual tracking [8,9]. However, the performance of deep learning models is depends on the quality of label annotations. These models are prone to memorizing noisy labels, which can lead to suboptimal learning outcomes [10]. Noisy labels are prevalent due to factors such as human errors and inaccuracies in labeling techniques, especially in large-scale datasets.

Learning with Noisy Labels (LNL) has thus become a crucial area in contemporary artificial intelligence research. The process typically begins with a "warm-up" stage where models are initially trained with noisy data. This stage is vital as it helps the model enhance its discriminative ability, enabling it to learn preferentially from correctly labeled data amidst noise. Nevertheless, this phase might inadvertently lead to the assimilation of incorrect information, with its efficiency being influenced by the noisy label, the number of training iterations, and specific model parameters.

As deep learning has advanced, a variety of models have been developed to mitigate the impact of noisy labels. Traditional approaches typically utilize CNNs or ViT. CNNs are constrained by their local receptive fields, limiting them to capturing only immediate spatial relationships. In contrast, ViT and its variants are proficient in constructing long-range dependencies due to their global receptive field, though they encounter increased computational complexity with larger input sizes. The Structured State-Space Sequence Model (S4) was first presented in [11], focusing on modeling long-range dependencies with

* Corresponding author.
E-mail addresses: ningweiwang@stu.xjtu.edu.cn (N. Wang), weiqiangjin@stu.xjtu.edu.cn (W. Jin), shirou_sav@berkeley.edu (S. Jing), haixia.bi@xjtu.edu.cn (H. Bi), g.yang@imperial.ac.uk (G. Yang).
[1] Both authors, N.W. and W.J., contributed equally to this work.

https://doi.org/10.1016/j.asoc.2024.112596
Received 23 June 2024; Received in revised form 20 November 2024; Accepted 5 December 2024
Available online 14 December 2024
1568-4946/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

linear complexity benefits. Building on this, several models have been developed, such as S5 [12], H3 [13], and GSS [14]. Moreover, pre-training on the ImageNet dataset with visual Mamba reduces external data dependency and noise memorization risk during the warm-up phase, leveraging a discriminative backbone network that effectively differentiates between correct and incorrect labels.

Additionally, two primary strategies are utilized to mitigate the impact of noisy labels [15,16]. The first strategy involves pre-training to identify and remove incorrect labels through methods such as manual inspection, clustering, or outlier detection algorithms [17,18]. Although effective, these methods often necessitate setting additional thresholds and can lead to the loss of informative data. Our methodology specifically targets noisy labels by refining predictions using softmax probabilities to gauge confidence levels, which then inform our entropy-variance label correction mechanism. We utilize entropy and variance as metrics to dynamically adjust the confidence threshold for label corrections, whereby predictions falling below a specified entropy-variance threshold trigger a review and potential correction of the corresponding labels.

The second strategy trains models to withstand noisy data by utilizing techniques that filter out noise, focusing on clean data subsets [19,20]. These techniques adjust the training focus based on model predictions to enhance handling of noise [21,22]. Despite the advancements of these techniques, the inherent bias towards easier classes in the selection process presents a challenge. This bias can lead to the neglect of complex yet informative samples, thus undermining the model's overall accuracy and effectiveness. To address this limitation and enhance the robustness of noise-handling in model training, our approach adopts a more nuanced and precise sample selection process that utilizes the feature space to assess sample proximity and relationships.

Considering the discussion of the above methods that prioritize reliable examples and ignore noise, which primarily focus on training with a clean subset of data, recent techniques present a shift in perspective. Recent techniques have shifted focus towards a semi-supervised learning paradigm, incorporating methods such as Mixup [23], MetaMixUp [24], and Mixup Inference [25]. These methods treat the selected clean subset of data as labeled samples and the non-selected subset as unlabeled, thus exploiting the entire dataset more effectively. However, these techniques often rely on fixed data augmentation strategies, which may not fully exploit the dynamic nature of training datasets. To overcome this limitation, our approach adjusts the intensity of data augmentation based on the variance of the data. It also innovatively rearranges parts of images in a puzzle-like manner, akin to Puzzlemix, to generate intricate composites, thereby further enriching the training process with a variety of challenging examples.

In this paper, we propose the Mamba and Entropy-variance Label Rectification (RelabelMamba) model to tackle the challenges outlined previously. The primary components of our model are detailed as follows:

- Vision Convolutional Mamba (Vcm): At the heart of the RelabelMamba model, the Vcm processes visual data through sequences of dimensionally enhanced patches embedded with positional information, which are managed within Vcm blocks. This structure allows our model to effectively integrate Mamba and label rectification techniques while maintaining high efficiency in the presence of noisy labels.
- Augment Entropy-variance K-Nearest Neighbors Relabel (AEKR): This component of our model combines Entropy-Variance Relabeling (ER) and K-Nearest Neighbors (KNN) to enhance training across various datasets. ER dynamically adjusts labels based on entropy and variance, reflecting the confidence in the model's predictions and adapting throughout the training phases to promote balanced and progressive learning. Concurrently, KNN leverages feature similarity and significance to select key samples, thereby enhancing learning efficiency and accuracy. Together, these methods counteract class bias by optimally balancing label correction and selective sample utilization, producing robust models that excel in LNL scenarios.
- Adaptpuzzle Mix: Incorporating Puzzlemix with Adaptivemix coefficient, Adaptpuzzle Mix enhances model robustness during training. Puzzlemix creates patches from two different augmentations of the same image, mixing them based on a probabilistic rule to blend features from both sources. The adaptive mixup coefficient, adjusted according to the variance of the data, linearly combines both inputs and their labels, enhancing the model's adaptability and effectiveness.

2. Related works

2.1. State space models

As deep learning evolves, a variety of models have been established as robust mechanisms for feature extraction during the warm-up phase. Traditional methods often employ CNNs [26–30] or ViT [31–35]. CNNs are celebrated for their scalability and straightforward computational complexity but are limited by their small receptive fields, resulting in a local bias. Furthermore, the uniform application of kernels across various input sections can reduce their adaptability in handling novel or low-quality images. Conversely, ViT methods enhance visual processing by using larger, global receptive fields and adaptable weights but face challenges due to the quadratic increase in computational complexity with larger input sizes. Efforts to optimize this, such as reducing dimensionality or adjusting stride in processing windows, often compromise the extent of receptive fields.

From the challenges faced by CNNs and ViT, Selective Structured State Space Models (SSM), often termed as Mamba, have emerged as an effective solution. Among these, the Mamba model [36] stands out by incorporating a data-dependent SSM layer and a selection mechanism that utilizes a parallel scan (S6), providing superior performance on large-scale datasets with linear scaling relative to sequence length. This configuration has been applied to various vision tasks including remote sensing imagery [37,38], medical image segmentation [39–41], image restoration [42], and point cloud analysis [43,44], consistently achieving competitive outcomes.

VMamba [45] extends the Mamba architecture to 2D image data, demonstrating enhanced performance in computer vision tasks by extending scanning in four directions to fully capture interrelations among image patches. MambaMorph [46], a novel derivative, combines Mamba models with CNNs to serve as a registration module in hybrid networks, accurately calculating deformation fields and showcasing the adaptability of Mamba models in enhancing traditional CNN architectures for complex vision tasks. U-Mamba [47] represents a hybrid CNN-SSM architecture designed to manage long-range dependencies in biomedical image segmentation. Vision Mamba [48] employs bidirectional SSMs for global context and includes position embeddings in its blocks to ensure precise visual recognition. It segments images into patches and uses bidirectional state space mechanisms to efficiently process large datasets and compress visual representations. These position embeddings enhance the model's spatial interpretation capabilities, boosting accuracy in noisy environments.

However, recent works have directly employed SSM as a plug-and-play module, lacking specific design adjustments for LNL challenges. Therefore, we propose that integrating Vision Mamba with CNNs in an attention-based fusion mechanism provides an effective solution to the issues presented by LNL. This Vision Mamba & CNNs hybrid model combines the global context handling capability of Vision Mamba with the hierarchical, local feature-capturing ability of CNNs. The synergy between these technologies allows the model to manage vast data volumes without sacrificing speed or accuracy.

Furthermore, a challenge associated with LNL is the tendency to produce inaccurate model representations, potentially affecting subsequent decision-making processes. This underscores the importance of investigating various LNL algorithms, typically categorized into three distinct approaches.

2.2. Relabeling noisy labels

MentorNet [20] utilizes a student network guided by pseudo-labels generated from a pre-trained network, providing a curriculum that focuses on reliable, less noisy examples to enhance learning quality. MetaCleaner [49] predicts confidence scores for noisy samples based on feature space proximity, using these scores to improve data representations and facilitate more effective learning from uncertain samples. Yi and Wu [50] advocate for the gradual adjustment of data labels based on network predictions, enabling continuous refinement throughout the learning process to better accommodate initial inaccuracies. Goldberger and Ben-Reuven [51] present a strategy using the Expectation–Maximization algorithm to model label noise and improve classifier performance by estimating correct labels. Wang et al. [52] address the challenge of applying differential learning rates to 'easy' and 'hard' classes through Symmetric Cross-Entropy Learning, which incorporates Reverse Cross-Entropy to enhance training robustness and prevent overfitting to specific error types. Han et al. [53] introduce a robust deep self-learning framework capable of effectively training on real noisy datasets without relying on assumptions about label noise distribution or requiring additional clean supervision. Research indicates that models trained on samples with noisy labels often converge more slowly than those trained with clean samples [54]. However, these models are vulnerable to performance degradation under conditions of high noise levels and a large number of classes, frequently requiring manual adjustments to thresholds for effective differentiation during various training epochs.

2.3. Selection from recovered labels

In recent times, the selection of samples within LNL has garnered increased attention. The primary distinction among these methods lies in the criteria for identifying clean samples. Early techniques were primarily focused on mitigating the impact of noisy samples by utilizing cleaner subsets during training [20]. Yu et al. [55] introduced a robust learning framework, Co-teaching+, which builds on the standard Co-teaching method by incorporating an "Update by Disagreement" strategy. This approach selectively back-propagates small-loss data from peer networks to enhance model robustness, demonstrating superior empirical results on benchmark datasets. Kim et al. [19] developed a method, SelNLPL, specifically aimed at addressing noisy labels in CNNs. Arpit et al. [54] proposed a selection strategy that leverages prediction likelihoods to segregate samples. Nevertheless, these strategies may encounter difficulties with highly complex noise patterns. Our proposed approach effectively addresses various types of label noise without adjustments to hyperparameters across different conditions, including diverse noise rates and types. Additionally, to balance the dataset and improve overall model performance, our technique strategically augments data categories with fewer corrected samples, ensuring equitable representation. This adjustment is essential for maintaining consistency during model training and improving the robustness of predictions across all categories.

2.4. Clean label usage

Previous techniques have aimed to mitigate the effects of noisy samples by selecting clean data subsets for training [20,21]. CutMix [56] alters patches between training images to enhance the model's tolerance to label noise by promoting robust feature learning. Similarly, Puzzlemix [57] shuffles parts of images in a puzzle-like fashion to create complex composites, thus potentially increasing the model's exposure to varied data without explicit reliance on label cleanliness. MixMatch [58] combines labeled and unlabeled data with a unified loss, effectively leveraging clean and noisy data by refining the labels during training. AugMix [59], developed by Hendrycks et al., introduces a simple data augmentation strategy that enhances model robustness and accuracy, particularly under distributional shifts in training and testing data that may resemble noise impacts. AutoMix by Liu et al. [60] fine-tunes the balance between mix policies and optimization complexity, proving particularly effective in environments with varying noise levels. The recent IPMix approach by Huang et al. [61] combines multi-level augmentations to expand data variety, enhancing both the precision and resilience of models in noisy settings without additional computational demands.

While these methods have made impressive strides, they often employ a static mixup strategy throughout the training process, which may not optimally address the variability in augmentation needs across different training stages. Intense data augmentation later in training can degrade performance by preventing the model from capturing subtle details in clean data samples. Moreover, earlier methods often fail to balance the use of clean and noisy data simultaneously. Although focusing on clean subsets can enhance model generalization, excluding noisy data too early may restrict the model's ability to learn from diverse data variations.

3. Methodology

Our approach starts with the introduction of necessary notations and proceeds with an elaborate explanation of the method we propose. The collection of labeled pairs is represented by $(\mathcal{X}, \mathcal{Y}) = (x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, where $\mathcal{X}$ symbolizes the data and $\mathcal{Y}$ denotes the associated labels. In this notation, each y_i is depicted as a one-hot encoded vector, indicating the class label among C potential classes.

3.1. Preliminaries

The primary focus of our study involves applying structured state space models (SSMs) such as the structured state space sequence models (S4) and the Mamba model to sequence data. These models are particularly suited for handling sequences where the input $x(t) \in \mathbb{R}^H$ and the output $y(t) \in \mathbb{R}$ are transformed using a hidden state $h(t) \in \mathbb{R}^N$. The evolution of the hidden state is governed by an evolution matrix $A \in \mathbb{R}^{N\times N}$ and projection matrices $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{1\times N}$:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t). \tag{1}$$

For sequence data analysis, we adapt these models to a discrete context with a timescale parameter Δ, using the zero-order hold (ZOH) method:

$$\bar{A} = \exp(\Delta A), \quad \bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B. \tag{2}$$

In the discrete model, the system equations modify to:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t. \tag{3}$$

The output is calculated via a global convolution, represented by:

$$K = (CB, CAB, \ldots, CA^{M-1}B), \quad y = x * K, \tag{4}$$

where M is the input sequence length, and K is a convolutional kernel.

3.2. Vision Convolutional Mamba (Vcm)

For vision applications, the Vision Convolutional Mamba (Vcm) model adapts the SSM framework to process 2-D images by segmenting each image into a sequence of 2-D patches. Each patch is flattened and projected to a higher dimension D, with the addition of positional embeddings E_{pos}:

$$T_0 = [t_{cls}; t_1^p W; t_2^p W; \ldots; t_J^p W] + E_{pos}. \tag{5}$$

The sequence undergoes processing through multiple Vcm blocks, each optimized for effective sequence modeling and handling bidirectional information flows:

$$T_l = \mathrm{Vcm}(T_{l-1}) + T_{l-1}, \quad f = \mathrm{Norm}(T_0^l). \tag{6}$$

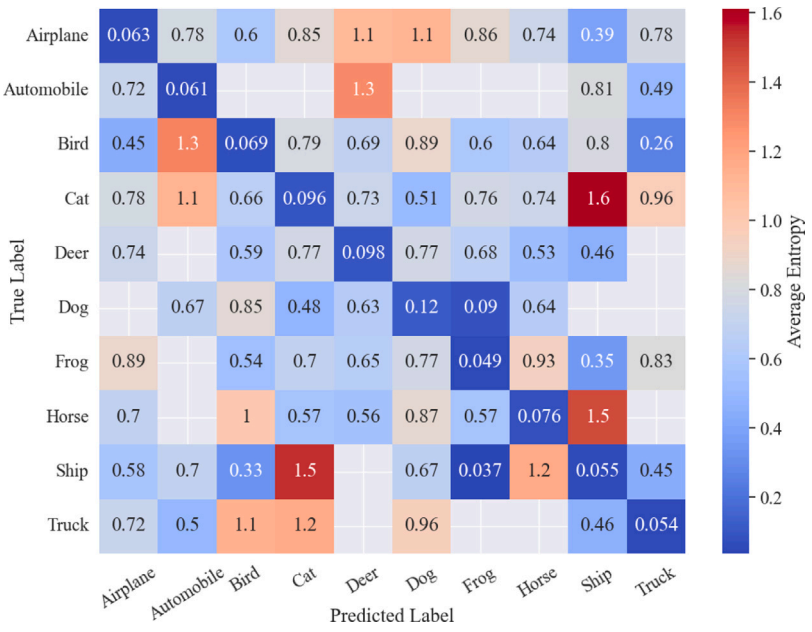


Fig. 1. Heatmap comparing the average entropy values between predicted and true labels. Darker shades of red indicate higher entropy values. The x-axis shows the predicted categories, while the y-axis represents the true categories.

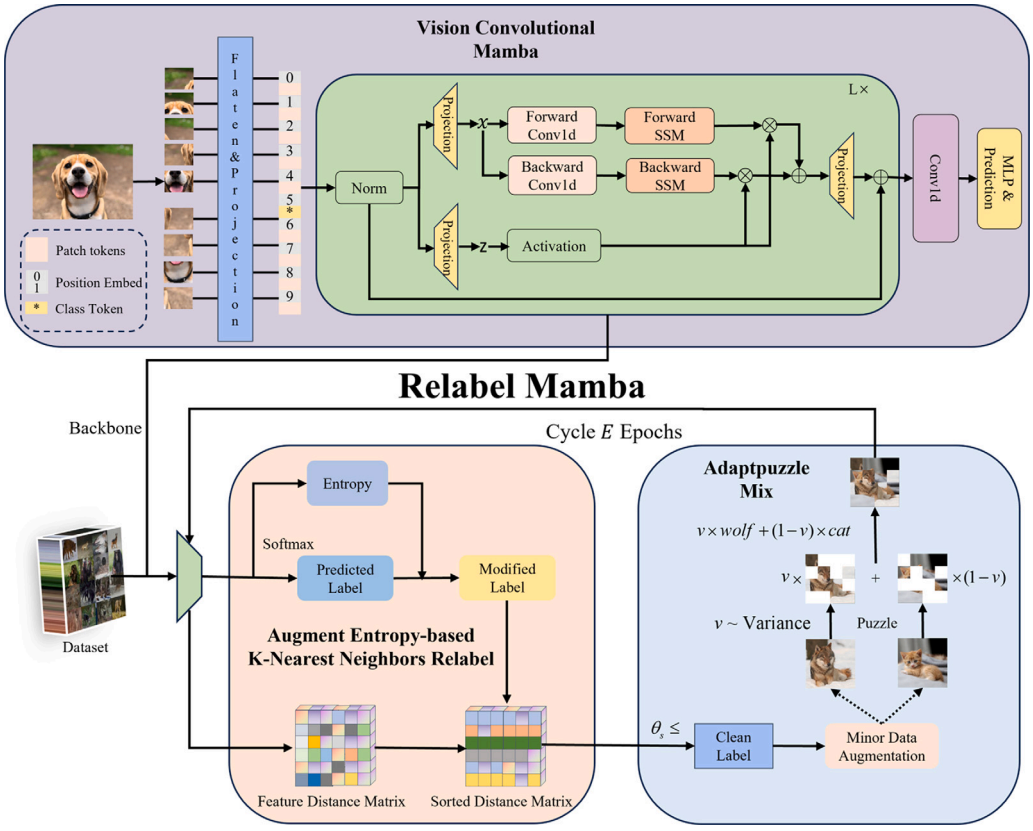


Fig. 2. Overview of our proposed RelabelMamba, which includes: Vcm to address the warm-up phase during pre-training; AEKR for noisy label correction, selection, and data augmentation; and the Adaptpuzzle method for utilizing soft labels.

Transformation within Vcm blocks involves linear projections and 1-D convolutions applied bidirectionally to manage spatial information efficiently. Finally, the classification output is derived from:

$$\hat{p} = \mathrm{MLP}(\mathrm{Conv1D}(f)). \tag{7}$$

The Vcm blocks incorporate unique features like linear projections and 1-D convolutions designed for bidirectional processing, enabling effective management of both forward and backward directional flows. The operations of the Vcm block are detailed in Fig. 2, where the input token sequence T_{l-1} is first normalized. This normalized sequence is then linearly projected into dimensions x and z of size E. The x

sequence is processed from both directions; for each direction, a 1-D convolution is applied to obtain x', which is then projected to B_o, C_o, and Δ_o. These are used to transform A_o and B_o. The outputs Y_{forward} and Y_{backward} from the sequence state model (SSM) are gated by z and combined to produce the final output token sequence T_1.

The architecture supports a variety of configurations, with parameters such as the number of blocks L, hidden state dimension D, expanded state dimension E, and SSM dimension N, optimized for different scales of the models, aligning with the standards set by ViT [62] and DeiT [63]. This unified approach underscores the adaptability of SSM-based models from sequence processing to LNL, leveraging structured transformations and deep learning paradigms.

3.3. Augment Entropy-variance K-nearest neighbors Relabel (AEKR)

Our proposed AEKR strategy improves label accuracy by combining an entropy-variance label correction mechanism with a k-Nearest Neighbors (KNN) relabeling approach. This methodology specifically targets samples in categories that demonstrate below-average data augmentation, thereby enhancing their classification accuracy. Initially, a Vcm pre-trained on ImageNet extracts and L_2 normalizes features from the input data $\mathcal{X}$, ensuring uniform feature magnitudes. The nearest neighbors for each sample are identified using a global distance matrix, with the most frequent label among these neighbors assigned as the corrected label. This method utilizes representation similarities to enhance classification accuracy for mislabeled samples.

In parallel, prediction scores generated by a classifier are normalized using softmax activation:

$$p(y_i|f_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}, \tag{8}$$

where $p(y_i|f_i)$ indicates the probability of label y_i given the feature vector f_i, and z_i is the logit corresponding to class i. To enhance the robustness of label correction, we define a threshold θ_h that combines both entropy and variance, capturing both the uncertainty and the consistency of predictions for each sample. The entropy $H(p)$ and variance $V(p)$ are both normalized to the $[0, 1]$ range to facilitate a balanced combination.

First, we compute the entropy for each sample:

$$H(p) = -\sum_{i=1}^{C} p(y_i|f_i)\log(p(y_i|f_i)), \tag{9}$$

where $H(p)$ reflects the uncertainty of the model's predictions. Higher entropy values indicate higher uncertainty. The heatmap in Fig. 1 illustrates the entropy distribution across predictions, with lighter areas showing where the model predicts with high confidence (low entropy) and darker areas indicating high uncertainty. Accurate predictions align with the lighter diagonal, while dark off-diagonal areas signal potential misclassifications. Predictions with notably low entropy, indicating high confidence, are used to rectify noisy labels, a method experimentally proven to be effective. Using the CIFAR-10 dataset as an example, as illustrated in Fig. 3, our approach improved label accuracy from 20% to an average of 96%, with some categories achieving up to 98%.

To ensure comparability across samples, $H(p)$ is normalized as follows:

$$\text{Norm}(H(p)) = \frac{H(p) - H_{\min}}{H_{\max} - H_{\min}}, \tag{10}$$

where $H_{\min}$ and $H_{\max}$ denote the minimum and maximum entropy values observed across the dataset, respectively.

Inspired by the Distinguishable Probability-Aware Distribution introduced in [64], which utilizes the difference between the top predicted probabilities to distinguish noisy labels, we define our threshold θ_h based on the difference between normalized entropy and variance. This approach allows us to capture samples that are uncertain yet inconsistent, which are more likely to be mislabeled. Similarly, we calculate the variance of the probability distribution for each sample, based on multiple predictions:

$$V(p) = \frac{1}{C}\sum_{i=1}^{C}\left(p(y_i|f_i) - \bar{p}\right)^2, \tag{11}$$

where $\bar{p} = \frac{1}{C}\sum_{i=1}^{C} p(y_i|f_i)$ represents the mean of the probabilities. $V(p)$ is then normalized:

$$\text{Norm}(V(p)) = \frac{V(p) - V_{\min}}{V_{\max} - V_{\min}}, \tag{12}$$

where $V_{\min}$ and $V_{\max}$ are the minimum and maximum variance values observed across the dataset. Our final threshold for label correction is defined as:

$$\theta_h = (\text{Norm}(H(p)) - \text{Norm}(V(p))) \times \frac{E_i}{E}, \tag{13}$$

Samples with θ_h below a dynamically adjusted threshold based on training progress are selected for relabeling, using the model's most confident predictions to enhance prediction reliability. This combined entropy-variance approach mitigates the risk of overfitting to noise by avoiding overcorrection of samples with stable uncertainty, which include difficult but correctly labeled instances.

For KNN, sample features are evaluated for proximity in feature space, with similarities expressed mathematically as:

$$\mathbf{s}_{ij} = \frac{f_i^T f_j}{|f_i|_2 \left|f_j\right|_2}. \tag{14}$$

These values are normalized and weighted to produce a score matrix that reflects the consensus between KNN-derived predictions and the revised labels. Clean samples, exhibiting marked consistency across prediction strategies, are identified and used to refine the model. This method also enhances data categories with fewer corrected samples, aligning them with the average. This integrative approach combines entropy-variance confidence assessment with the robust, discriminative capabilities of KNN, effectively improving label accuracy and model robustness against noise.

3.4. Adaptpuzzle coefficient

Incorporating Puzzlemix [57] with the novel Adaptivemix coefficient, Adaptpuzzle Mix enhances model robustness during training. These methods synergistically enhance model robustness by dynamically manipulating input data based on feature variance. Puzzlemix enhances training by promoting model robustness through local perturbations. It utilizes data from two input images x_a and x_b. The function divides each input tensor into a grid of patches and randomly interchanges these patches based on a simple probabilistic rule:

1. Divide each input tensor into $n_s \times n_s$ patches.
2. For each patch position, randomly decide whether to keep the patch or replace it with a corresponding one. This decision is made using a random matrix M where M_{ij} is a Bernoulli random variable: $M_{ij} \sim \text{Bernoulli}(\upsilon)$.
3. Reconstruct the mixed input by combining the selected patches from both inputs, maintaining their original spatial order.

The final mixed image x_{mix} is reconstructed by assembling these patches, creating inputs that are a blend of features from both source images.

To dynamically control the augmentation intensity, we define an adaptive coefficient υ that is sensitive to the variance in feature space between two samples, x_a and x_b. The coefficient υ is calculated as follows:

1. Variance Calculation: First, we compute the variance $V(x_a)$ and $V(x_b)$ for each sample individually.

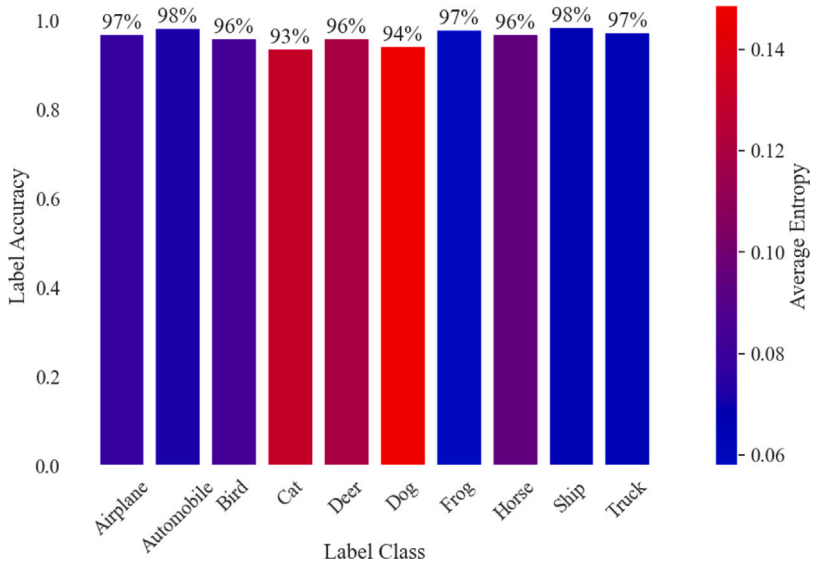


Fig. 3. The chart shows label rectification accuracy by class, with darker blue bars indicating higher model certainty and lower entropy.

2. Proportion Determination: Using the variances, we calculate the proportion of variance contributed by x_a relative to the combined variance, defined as:
$$\upsilon = \frac{V(x_a)}{V(x_a) + V(x_b)}.$$

For two inputs x_a and x_b with their associated targets y_a and y_b, the mixed input and target are derived as:

$$x_{mix} = Puzzlemix(x_a, x_b, \upsilon). \tag{15}$$

$$y_{mix} = \upsilon \times y_a + (1 - \upsilon) \times y_b. \tag{16}$$

The model is then trained on these mixed inputs and targets, optimizing a loss function that encourages the model to predict the blended labels accurately, thereby enhancing generalization and robustness. For each pair x_{mix}, y_{mix} in the selected subset $(\mathcal{X}_{mix}, \mathcal{Y}_{mix})$, the encoder f utilizes a standard cross-entropy loss, formulated as: $L_{ce} = -\boldsymbol{y_{mix}}^T \log f(\boldsymbol{x_{mix}})$.

4. Experiments

This manuscript conducts an exhaustive evaluation of the introduced approach through both quantitative assessments and qualitative reviews across four distinct datasets: CIFAR-10/100, Clothing1M, and ANIMAL-10N. The experimental framework, including detailed descriptions of the datasets and the settings of the parameters, is thoroughly elaborated upon in Sections 4.1 and 4.2. The efficacy of key components of our model is showcased in the ablation study in Section 4.3. This study features entropy-variance label rectification, a KNN-based selection of clean samples with augmentation, the use of Adaptpuzzle mixup for soft labeling in the dataset, and a pre-trained backbone based on vision convolutional networks.

4.1. Datasets

Our evaluation utilizes the CIFAR-10 and CIFAR-100 datasets, crafted by Krizhevsky and colleagues [65], consisting of a total of 60,000 images divided into distinct categories. CIFAR-10 contains 10 classes, while CIFAR-100 is divided into 100 classes grouped into 20 superclasses. We apply both symmetric and asymmetric synthetic noise models to test the robustness against noise. Asymmetric noise introduces disturbances that vary in magnitude and characteristics across different transmission directions, causing non-uniform impacts and potential mislabeling, especially noticeable in CIFAR-100's superclass-related label shifts. In contrast, symmetric noise impacts the data uniformly, presenting equal levels of disturbance across all directions. For symmetric and asymmetric noise generation, we followed standard protocols as described in related literature [21,22].

The Clothing1M dataset, created by Xiao and colleagues [16], comprises 1 million noisy images from 14 different clothing categories gathered from e-commerce sites. Mislabeling is prevalent, influenced by inaccurate seller descriptions, especially in similar categories like Knitwear and Sweater, with an estimated noise ratio of 38.5%. The ANIMAL-10N dataset, introduced by Song et al. [66], contains 50,000 training and 10,000 test images across 10 animal classes, with an 8% estimated error rate due to manual labeling inaccuracies.

4.2. Training details

The Vcm network initiates training using weights that are pre-trained on ImageNet to leverage foundational visual features beneficial for generalization across different datasets. Subsequently, the further training on CIFAR-10/100 and ANIMAL-10N proceeds with an initial learning rate of 0.0002 and a momentum of 0.9, using a batch size of 22 over a compact 50-epoch schedule. This combination allows for efficient convergence, as the pre-trained weights facilitate rapid adaptation, reducing the training time while achieving high accuracy.

For the Clothing1M dataset, we utilize the ResNet50 model, which also incorporates pre-trained ImageNet weights as recommended by Li et al. [22]. The training spans 150 epochs with an initial learning rate of 0.002, which is reduced after the 50th and 100th epochs. The model is trained using a batch size of 32.

4.3. Ablation study

To evaluate our method for correcting noisy labels, we conducted experiments on CIFAR-10 using uniform conditions: 50% symmetric noise, 50 epochs, and the Vcm pre-trained backbone. We divided the experiments into four distinct groups, each designed to highlight specific features of our proposed model:

- **SSR+**: Serving as the baseline, this group employs a SOTA correction approach without modifications derived from our study.
- **Vcm**: This group evaluates the Vcm model's effectiveness by training it directly on a noisy dataset without label correction mechanisms.

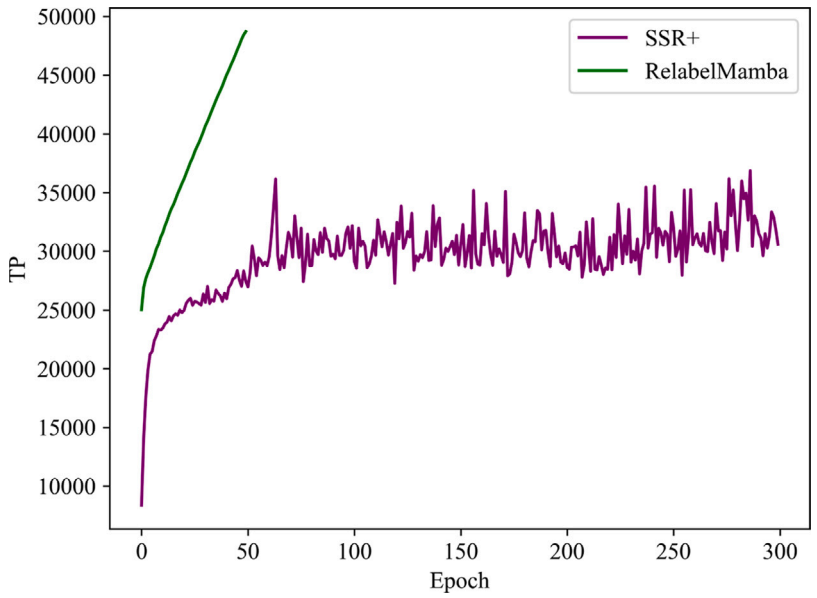


Fig. 4. Comparative analysis of label refine performance between RelabelMamba over 50 epochs and SSR+ over 300 epochs.

Table 1
Performance comparison of methods in terms of test accuracy and True Positives (TP).

Method	Test accuracy	TP
SSR+	96.1	30,600
Vcm	86.9	–
Vcm+AEKR	96.3	45,326
RelabelMamba	**96.8**	**48,719**

- **Vcm+AEKR**: An advanced variant of the Vcm model, this setup integrates a Augment Entropy-Variance K-Nearest Neighbors Relabel (AEKR) approach with data augmentation strategies to improve label accuracy and enhance the model's robustness against noise.
- **RelabelMamba**: This variant combines Mamba label rectification with Puzzlemix and Adaptivemix coefficient.

4.3.1. Vision Convolutional Mamba (Vcm)

SSR+, serving as the baseline, employs a SOTA correction method, setting a high benchmark with a test accuracy of 96.1% and 30,600 true positives, as shown in Table 1. The robustness of this model against noise illustrates the effectiveness of conventional noise correction strategies in maintaining high performance. In contrast, the Vcm incorporates a Mamba model pre-trained on the ImageNet dataset, which is optimized for efficient pre-training on labeled data.

However, without specific noise correction adaptations, the Vcm model achieves a test accuracy of 86.9%. Although lower than SSR+, this performance provides insights into the intrinsic capabilities of the Vcm framework under noisy conditions, highlighting the model's vulnerability to noise and the necessity for integrated label correction mechanisms to enhance its robustness. The complex architecture of Vcm captures detailed features and dependencies within the data, offering a theoretical advantage in learning from both structured and unstructured environments. By focusing on inherent structures and regularities in the input data, Vcm enhances robustness and transferability of representations, minimizes noise memorization, and facilitates a smooth warm-up phase. This promotes training stability, reduces overfitting, and enhances overall performance.

However, applying Vcm directly to a noisy dataset without tailored enhancements reveals that while the model is technically advanced, it still requires further refinement to effectively handle the practical challenges of label noise. The substantial performance gap between Vcm and SSR+ underscores the importance of integrating effective noise management strategies within the Vcm training protocol.

4.3.2. Augment Entropy-variance K-nearest neighbors Relabel (AEKR)

The Vcm+AEKR setup demonstrates a considerable enhancement in handling noisy labels, utilizing Vcm model in conjunction with an AEKR approach. This integration aims to specifically address the vulnerabilities exposed in the Vcm-only group, by incorporating robust label correction mechanisms directly into the model's training process. The primary advantage of the Vcm+AEKR approach is its sophisticated label correction mechanism. It employs an entropy-variance method to evaluate prediction certainty and a KNN strategy to refine labels according to the most common neighboring class, effectively mitigating the effects of noisy labels. The entropy and variance measures allow the model to identify and focus on samples with predictable labels, while KNN leverages spatial coherence in the feature space to adjust uncertain labels towards their more probable classes. This dual strategy not only improves the accuracy of the label set but also enhances the model's ability to train on accurate data, thereby increasing learning efficacy.

The integration of AEKR within the Vcm framework also contributes to increased robustness against label noise. By selectively correcting labels that are likely erroneous before they can mislead the training process, Vcm+AEKR prevents the accumulation of errors that could otherwise result from training on mislabeled data. This preemptive correction maintain the integrity of the model's learning path, promoting more reliable and stable generalization across different datasets. As indicated in the performance metrics in Table 1, Vcm+AEKR achieves a test accuracy of 96.3% with a noteworthy number of true positives at 45,326. This represents a notable improvement over the standalone Vcm model, which had lower accuracy and struggled with noisy conditions. The enhanced performance highlights the effectiveness of the AEKR mechanism in refining training data quality, thereby enabling the Vcm model to fully exploit its sophisticated architecture.

4.3.3. RelabelMamba

The Adaptpuzzle Coefficient, integral to our method, enhances performance by dynamically adjusting the blending coefficient based on training progression. This strategy prevents the model from overfitting

Table 2
Performance and computational cost of KNN-based label correction for different values of k on CIFAR-10 and CIFAR-100 datasets.

Dataset	Noise type	Noise rate	k	Accuracy (%)	Time (s)
CIFAR-10	Symmetric	50%	50	97.0	0.34
			100	96.9	0.39
			200	96.8	0.46
			300	96.9	0.52
			500	96.9	0.61
CIFAR-100	Asymmetric	30%	50	83.5	0.53
			100	82.9	0.62
			200	82.6	0.75
			300	82.9	0.94
			500	82.4	1.28

to noise, fostering a more generalized data representation. Such adaptability is vital in managing the uncertainty associated with noisy labels during training.

RelabelMamba exhibits remarkable efficiency in learning and precision in identifying true positives, achieving substantial results within a reduced training period.

As shown in Fig. 4, SSR+ experiences a rapid rise in true positives during the initial phase, reaching a plateau around 100 epochs, with minimal gains beyond that point, even with continued training up to 300 epochs. However, after peaking early in the training process, the performance plateaus. This suggests that SSR+ quickly learns to identify true positives but does not noticeably improve beyond the initial learning phase, even with extended training up to 300 epochs. In contrast, RelabelMamba, at the end of the first epoch, reaches 25,036 true positives, whereas SSR+ only achieves 8368 true positives at the same point. It maintains or improves performance with fewer epochs, reducing computational resources and training time compared to traditional methods like SSR+.

The quantitative data in Table 1 reveals that RelabelMamba achieves a test accuracy of 96.8% and records 48,719 true positives, marking substantial improvements over the baseline SSR+ model, which scores a test accuracy of 96.1% with 30,600 true positives. This highlights the efficacy of integrating Mamba label rectification with advanced data augmentation techniques, such as Puzzlemix and Adaptivemix strategy.

4.3.4. Acceleration strategy for KNN-based label correction

To address the computational challenges posed by the KNN-based label correction strategy in AEKR, we propose a comprehensive acceleration strategy that enhances efficiency without compromising accuracy. This strategy integrates chunk-based processing, feature normalization, weighted similarity calculation, and dynamic score adjustment, ensuring practical applicability to large-scale datasets. The KNN algorithm's complexity, driven by the need to compute pairwise distances across all samples in the dataset, can be alleviated through a chunk-based computation approach. By dividing the dataset into smaller, manageable chunks, we process each subset independently, identifying nearest neighbors locally before aggregating results across chunks. This division reduces memory usage, as the entire dataset does not need to be loaded simultaneously, and accelerates computation by enabling parallel processing. In addition to chunk-based processing, feature vectors are normalized to ensure uniform scaling across the feature space. By applying L_2-normalization, the magnitude of all feature vectors is standardized, ensuring that each feature contributes equally to the distance calculations. This normalization not only enhances the consistency of similarity measures but also reduces the variability in feature distributions, improving computational efficiency and robustness in noisy environments.

We also introduce a weighted similarity calculation to further optimize the label correction process. During the KNN step, pairwise similarities are scaled using adaptive weights that prioritize the most relevant neighbors, thereby refining the corrected labels. The weights are normalized dynamically within each chunk, maintaining consistency across the entire dataset. By focusing on influential neighbors, this strategy reduces the overall computational burden without compromising the effectiveness of the correction process. After aggregating the similarity scores from all chunks, a dynamic score adjustment is applied to ensure coherence in the final predictions. This adjustment globally normalizes the scores, preserving the integrity of the corrected labels across the dataset. By scaling predictions dynamically, the algorithm maintains a coherent prediction space, even when processed in segments, enabling accurate and efficient label correction.

The efficacy of this acceleration strategy was validated through experiments on CIFAR-10 and CIFAR-100, as summarized in Table 2. The results demonstrate that the computational burden of KNN can be significantly reduced without sacrificing accuracy. For instance, on CIFAR-10, increasing k from 50 to 500 led to only a marginal increase in runtime (from 0.34 s to 0.61 s), while accuracy remained consistently above 96.9%. A similar trend was observed on CIFAR-100, where the runtime increased from 0.53 s to 1.28 s for the same k-range, with accuracy stabilizing around 83.5%. These findings highlight that chunk-based processing, normalization, and weighted similarity adjustments together effectively mitigate the computational cost traditionally associated with KNN-based methods. This acceleration strategy ensures that AEKR can be efficiently applied to large-scale datasets while maintaining its robustness and effectiveness.

5. Results

In this section, we present the performance of our RelabelMamba method under various label noise conditions, starting with synthetic datasets such as CIFAR-10/100 and progressing to real-world datasets like ANIMAL-10N and Clothing1M.

5.1. Symmetric noise

Symmetric noise assumes that each label can be randomly flipped to any other class label with equal probability, mimicking scenarios where annotation errors are uniformly distributed across all classes. Mathematically, if the original label of a sample is y_i, it will remain unchanged with probability $1-\eta$. However, with probability η, the label y_i is randomly changed to any of the remaining $C-1$ class labels. The noise transition matrix T for symmetric noise is represented as:

$$T_{ij} = \begin{cases} 1-\eta & \text{if} \quad i = j \\ \frac{\eta}{C-1} & \text{if} \quad i \neq j, \end{cases}$$

where C is the total number of classes, and η is the noise ratio (e.g., 20%, 50%). In this scenario, the impact of label noise is evenly distributed, presenting a uniform level of difficulty across all classes.

We evaluated the RelabelMamba method under different levels of symmetric noise in the CIFAR-10 and CIFAR-100 datasets, demonstrating noticeable improvements over existing techniques. Table 3 summarizes these results, highlighting our method's superiority across various noise intensities. For CIFAR-10, under 20%, 50%, and 80% noise levels, RelabelMamba achieves accuracies of 97.0%, 96.8%, and 96.2% respectively. In comparison, the nearest competitor, which achieves an accuracy of 95.6% under 80% noise, our method shows a improvement of 0.6%. This illustrates the distinct advantage of RelabelMamba, especially in environments with substantial symmetric noise, where it exceeds many other existing models. Similarly, for CIFAR-100, RelabelMamba leads with accuracies of 83.2%, 80.5%, and 75.8% across the 20%, 50%, and 80% noise levels. It demonstrates less performance degradation as noise increases, maintaining a more stable performance curve. The decrease in performance is less than 8% even at 80% noise, demonstrating its robustness compared to other models that experience sharp drops at higher noise levels.

Table 3
Test accuracies (%) obtained by different techniques under symmetric and asymmetric noise on CIFAR-10 and CIFAR-100 datasets. The best results are in bold. The data are copied from respective papers. The - denotes the lack of respective data.

Method	Symmetric noise						Asymmetric noise					
	CIFAR-10			CIFAR-100			CIFAR-10			CIFAR-100		
	20%	50%	80%	20%	50%	80%	10%	30%	40%	10%	30%	40%
CE	86.8	79.4	62.9	62.0	46.7	19.9	88.8	81.7	76.1	68.1	53.3	44.5
LDMI [67]	88.3	81.2	43.7	58.8	51.8	27.9	91.1	91.2	84.0	68.1	54.1	46.2
TCC-net [68]	91.9	89.3	–	68.9	62.4	–	–	89.2	82.7	–	62.9	54.6
TAkS [69]	83.9	76.8	–	50.7	41.0	–	–	–	73.4	–	–	35.2
PCIL [50]	92.4	89.1	77.5	69.4	57.5	31.1	93.1	92.9	91.6	76.0	59.3	48.3
JPL [70]	93.5	90.2	35.7	70.9	67.7	17.8	92.0	92.5	90.7	72.0	68.1	
UNICON+HMW [71]	93.5	95.2	93.7	76.6	75.8	63.4	93.5	94.7	93.7	76.7	76.3	72.1
[71] MOIT [72]	94.1	91.1	75.8	75.9	70.1	51.4	94.2	94.1	93.2	77.4	75.1	74.0
Robust DivideMix [73]	95.5	–	61.6	77.4	–	48.1	–	–	86.9	–	–	–
M-Up [23]	95.6	87.1	71.6	67.8	57.3	30.8	93.3	83.3	77.7	72.4	57.6	48.1
CRAS [74]	95.7	95.4	93.8	79.5	77.0	65.0	–	–	93.1	–	–	78.0
RRL [75]	95.8	94.3	92.4	79.1	74.8	57.7	–	–	91.9	–	–	–
ELR+* [76]	95.8	94.8	93.3	77.6	73.6	60.8	95.4	94.7	93.0	77.3	74.6	73.2
NGC [77]	95.9	94.5	91.6	79.3	75.9	62.7	–	–	90.6	–	–	–
TCL+* [78]	96.0	94.5	–	79.3	74.6	–	95.9	94.8	93.7	79.0	76.9	–
UniCon* [79]	96.0	95.6	93.9	78.9	77.6	63.9	95.3	94.8	94.1	78.2	75.6	74.8
OT-Filter [80]	96.0	95.3	94.0	76.7	73.8	61.8	–	–	95.1	–	–	76.5
DivideMix* [22]	96.1	94.6	93.2	77.3	74.6	60.2	93.8	92.8	91.7	71.6	69.5	55.1
LongReMix [81]	96.3	95.1	–	77.9	75.5	–	–	–	94.7	–	–	59.8
LNL-FG [82]	–	95.2	–	–	74.7	–	–	–	94.9	–	–	75.1
DMLP-DivideMix [83]	96.3	95.8	94.5	79.9	76.8	68.6	–	–	95.0	–	–	–
AugDesc* [84]	96.3	95.4	93.8	79.5	77.2	66.4	–	–	94.6	–	–	–
DSCL+ [85]	96.3	95.5	93.8	79.5	76.7	70.3	96.8	95.2	92.9	80.1	77.9	76.1
C2D* [86]	96.4	95.3	94.4	78.7	76.4	67.8	–	–	93.5	–	–	–
Robust LR* [87]	96.5	95.8	94.3	79.1	75.3	66.7	–	–	94.4	–	–	–
RML-Semi [88]	96.5	95.7	93.9	78.9	77.8	64.1	–	–	–	–	–	–
DPC* [89]	96.5	95.9	94.8	81.0	78.5	66.4	95.4	95.3	94.9	80.5	79.7	75.6
InstanceGM [90]	96.7	95.9	–	79.7	77.2	–	–	–	–	–	–	–
SSR+ [91]	96.7	96.1	95.6	79.7	77.2	–	–	–	95.5	–	–	–
RegCE+semi [92]	96.8	96.0	93.5	80.5	77.3	66.4	–	–	92.2	–	–	66.8
RelabelMamba (ResNet18)	96.8	96.5	95.7	81.1	78.7	69.6	96.5	95.9	95.7	81.1	80.0	78.6
RelabelMamba (ViT)	96.9	96.7	95.9	82.6	80.3	72.1	96.8	96.4	96.0	82.3	81.5	79.2
RelabelMamba (Vcm)	**97.0**	**96.8**	**96.2**	**83.2**	**80.5**	**75.8**	**97.0**	**96.7**	**96.4**	**83.5**	**82.8**	**82.0**

The superior handling of symmetric noise by RelabelMamba can be attributed to its integration of advanced label correction techniques, including Mamba label rectification and the AEKR strategy, which effectively mitigate the effects of mislabeled data. The model's dynamic data augmentation strategies, such as Puzzlemix and an Adaptivemix coefficient, allow it to adapt to variability and uncertainty, avoiding overfitting to noisy labels. Its robust design continues to refine training data and labels, maintaining high accuracy across various scenarios. Further statistical analysis could be conducted to validate the significance of these performance differences.

To evaluate the effectiveness of integrating the AEKR strategy across different architectures, we conducted a comparative study involving ResNet18, ViT, and Mamba (Vcm). Each architecture was enhanced with AEKR, and the results are summarized in Table 3. This comparison highlights the advantages of the Mamba architecture when paired with AEKR. The ResNet18 architecture combined with AEKR achieved a test accuracy of 96.8% on CIFAR-10 and 81.1% on CIFAR-100. Its hierarchical feature extraction enabled effective noise management in simpler datasets like CIFAR-10. However, in CIFAR-100, its smaller receptive field restricted its ability to model long-range dependencies, resulting in an accuracy drop of 2.1% compared to Mamba (Vcm). This limitation was particularly evident in class granularity, where the noise from closely related classes impacted performance.

ViT, inspired by [93] and integrated into our AEKR framework, exhibited better performance than ResNet18, achieving 96.9% on CIFAR-10 and 82.6% on CIFAR-100. On CIFAR-100, ViT benefited from its global attention mechanism, yielding an improvement of 1.5% over ResNet18, especially in classes with interdependent features. Mamba (Vcm) with AEKR consistently outperformed the other architectures, achieving the highest test accuracy of 97.0% on CIFAR-10, 83.2% on CIFAR-100. On CIFAR-100, its performance exceeded ResNet18 by 2.1% and ViT by 0.6%, demonstrating its superior handling of noisy, high-class diversity datasets. This study provides a direct comparison of ResNet18, ViT, and Mamba (Vcm) when integrated with AEKR. The results emphasize the advantages of the Mamba architecture, which consistently delivered superior performance by leveraging its structured state space model, efficient noise correction, and scalable feature extraction capabilities. These findings establish Mamba (Vcm) as the preferred architecture for robust learning under noisy conditions.

5.2. Asymmetric noise

Asymmetric noise, in contrast, reflects a more realistic scenario where errors occur in a biased manner—labels are more likely to be confused with certain similar classes. This type of noise is especially relevant for datasets with visually similar categories. The noise transition matrix T for asymmetric noise is crafted based on class-specific similarities. For example, in CIFAR-10, cats may be misclassified as dogs, and trucks may be mislabeled as automobiles. Similarly, in CIFAR-100, which contains 100 classes grouped into 20 superclasses, the noise is introduced in a way that label shifts occur within the same superclass. Formally:

$$T_{ij} = \begin{cases} 1-\eta & \text{if} \quad i = j \\ \eta & \text{if} \quad (i,j) \in \text{confusing pairs} \\ 0 & \text{otherwise.} \end{cases}$$

This matrix emphasizes targeted label noise, presenting a non-uniform challenge for the learning algorithm. Asymmetric noise is particularly useful for assessing a model's ability to handle domain-specific or semantic misclassifications.

The performance of the RelabelMamba method under asymmetric noise conditions further highlights its robustness and superior handling

Table 4
Testing accuracy (%) on ANIMAL-10N.

Method	Test accuracy
CE	79.4
SELFIE [66]	81.8
TCC-net [68]	83.2
PLC [94]	83.4
NCT [95]	84.1
InstanceGM [90]	84.6
OT-Filter [80]	85.5
SLRLNL [96]	86.4
CCE+HMW [71]	86.5
Dynamic Loss [97]	86.6
DISC [98]	87.1
RegCE [92]	88.3
SSR+ [91]	88.5
Robust LR [87]	88.5
Jigsaw-Vit [93]	89.0
SURE [99]	89.0
CWCL [100]	89.0
PASS [101]	89.2
SANM [83]	89.3
DivideMix+MC+SDA [102]	90.5
RelabelMamba (ours)	**92.9**

Table 5
Testing accuracy (%) on Clothing1M.

Method	Test accuracy
CE	69.21
TCC-net [68]	70.46
TAkS [69]	70.28
SLRLNL [96]	74.15
RRL [75]	74.30
C2D* [86]	74.84
ScanMix [103]	74.35
DivideMix* [22]	74.76
LongReMix [81]	74.38
InstanceGM [90]	74.40
OT-Filter [80]	74.50
CRAS [74]	74.54
DIST+DivideMix [98]	74.79
TCL+* [78]	74.85
ELR+* [76]	74.81
SSR+ [91]	74.83
UNICON* [79]	74.98
AugDesc* [84]	75.11
RML-Semi [88]	75.14
DivideMix*+SNSCL [82]	**75.31**
RelabelMamba (ours)	74.98

of label inaccuracies. The results presented in Table 3 demonstrate our method's capability to outperform SOTA models under varying degrees of asymmetric noise on both CIFAR-10 and CIFAR-100 datasets. For CIFAR-10, RelabelMamba achieves impressive accuracies of 97.0%, 96.7%, and 96.4% at noise levels of 10%, 30%, and 40%, respectively. These results consistently surpass those of the nearest competitors, most notably at the 40% noise level, where RelabelMamba leads by 0.9%. This margin illustrates the effectiveness of our method's adaptive mechanisms, which excel in environments with skewed label distributions that often challenge traditional learning models.

Similarly, in the CIFAR-100 dataset, our method achieves leading accuracies of 83.5%, 82.8%, and 82.0% across the same noise levels. This performance is remarkable, considering the complexity and higher class count of CIFAR-100, which typically exacerbates the challenges posed by asymmetric noise. The stability and reliability of RelabelMamba are evident as it maintains less than a 2% drop between 10% and 40% noise levels, showcasing its ability to preserve learning integrity despite notable label noise.

Fig. 5 features T-SNE visualizations of network features from CIFAR-10 dataset test images after 50 epochs of training under different label noise conditions: symmetric at 20% and 60%, and asymmetric at 20% and 60%. Each plot displays clusters of points, each color-coded to represent one of the ten CIFAR-10 classes—Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck. Fig. 5(a)-Symmetric 20% shows distinct class clusters with minimal overlap even at 20% noise, highlighting RelabelMamba's ability to maintain accurate class distinctions. Despite higher noise, Fig. 5(b)-Symmetric 60% features distinctly clustered points with slight blending, demonstrating the robustness of the model's noise mitigation strategies. Fig. 5(c)-Asymmetric 20% maintains clear class separation with minor overlaps, even with moderate asymmetric noise, indicating strong feature extraction capabilities. Lastly, Fig. 5(d)-Asymmetric 60% displays more mixed clusters than at 20% noise but still retains appreciable class separation, showcasing the model's capacity to handle severe noise and preserve class integrity effectively.

5.3. ANIMAL-10N

The ANIMAL-10N dataset, renowned for its noisy labels that closely mimic real-world conditions, presents considerable challenges that test the noise handling robustness of learning models. The evaluation of our RelabelMamba method on the ANIMAL-10N dataset further substantiates its efficacy in handling real-world label noise conditions. As indicated in Table 4, RelabelMamba surpasses various recent methods, achieving a remarkable test accuracy of 92.9%. This demonstrates a substantial improvement over the nearest competing methodologies, such as DivideMix+MC+SDA, which has an accuracy of 90.5

The superior performance of RelabelMamba in this context is a testament to its advanced label noise correction and robust feature learning strategies. The impressive accuracy achieved on ANIMAL-10N highlights RelabelMamba's capability to adapt and perform robustly across real-world label noise, making it an ideal choice for practical applications where label quality is uncertain. This performance reflects the method's design philosophy, which prioritizes adaptability, robustness, and the ability to learn true class distributions from corrupted labels.

5.4. Clothing1M

Clothing1M is widely recognized for its substantial real-world label noise, establishing it as a crucial benchmark for assessing noise robustness in LNL. The dataset comprises one million clothing images with labels automatically generated from surrounding text, resulting in a high rate of incorrect labels. This presents a unique challenge, as the noise is neither entirely symmetric nor asymmetric but rather a complex mix that resembles real-world data errors.

The evaluation of the RelabelMamba method on the real-world Clothing1M dataset demonstrates its robustness and efficacy in handling label noise in practical scenarios. As shown in Table 5, RelabelMamba achieves a accuracy of 74.98%, which places it among the top performers on this challenging dataset. RelabelMamba's performance on Clothing1M is impressive, especially when compared to other SOTA methods. Although it does not reach the highest accuracy – being marginally surpassed by DivideMix*+SNSCL at 75.31% – its performance is impressively close, within a 0.33% gap of the leading score. This narrow discrepancy highlights the effectiveness of RelabelMamba's approach to managing diverse and unpredictable noise patterns, a crucial factor for real-world applications. Furthermore, the slight underperformance of RelabelMamba compared to leading models might stem from the distinct noise characteristics in Clothing1M, suggesting the need for further refinements in label correction techniques or training strategies. Future work should focus on optimizing these aspects for the types of noise prevalent in datasets like Clothing1M. Such enhancements could improve accuracy and position RelabelMamba as the preferred option for similar real-world applications, emphasizing the need for continual adaptation and refinement in developing noise-robust machine learning models for diverse, noise-rich environments.

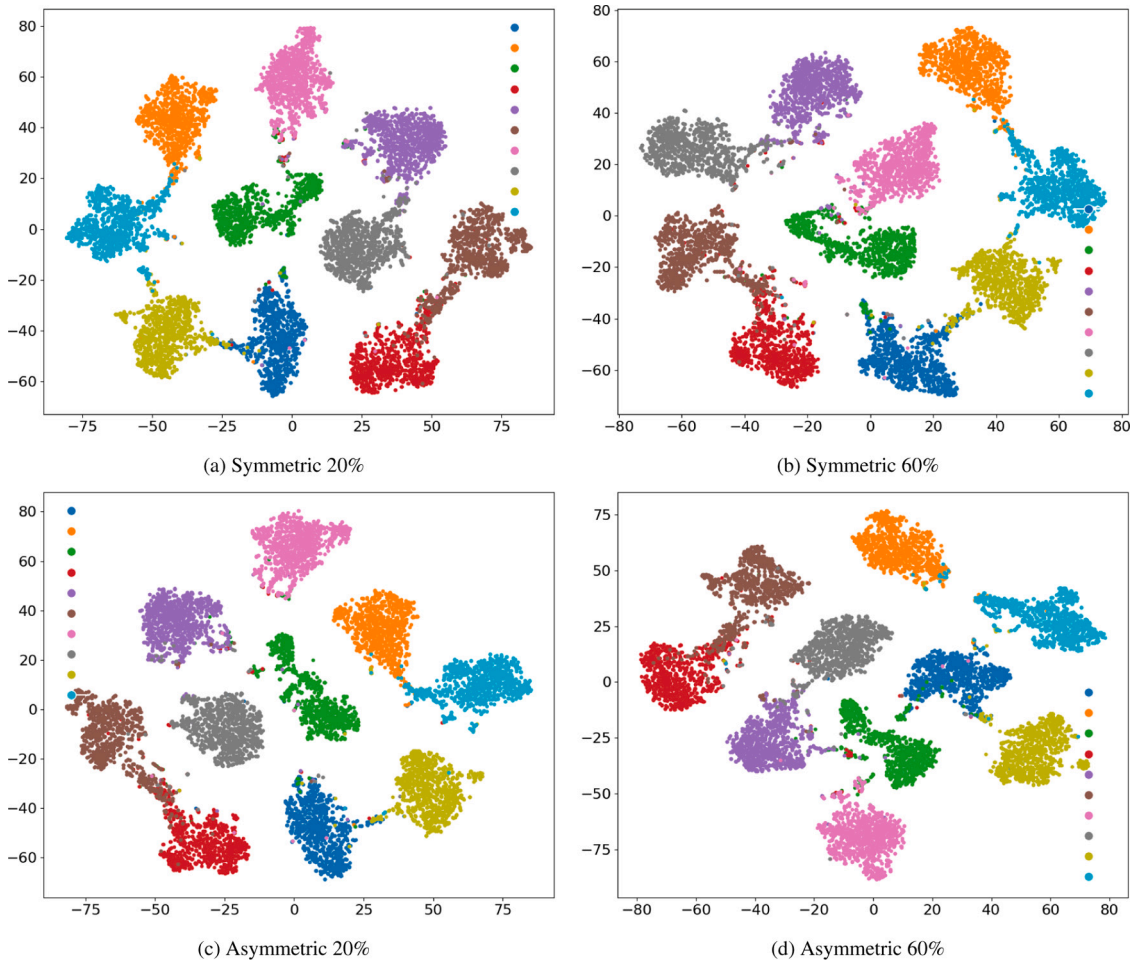


Fig. 5. The T-SNE visualizations provide insights into the network features of test images from the CIFAR-10 dataset, depicting the class distribution after 50 epochs of training with various noise types: symmetric at 20% and 60%, and asymmetric at 20% and 60%. Despite the challenges of extreme label noise, RelabelMamba's capability is evident in its effective learning of true class distributions. Each of the ten classes—*Airplane*, *Automobile*, *Bird*, *Cat*, *Deer*, *Dog*, *Frog*, *Horse*, *Ship*, *Truck*—is distinctly represented by a unique color: deep blue, vibrant orange, grassy green, bold red, muted purple, earthy brown, soft pink, medium gray, bright lime green, and clear aqua, respectively.

6. Conclusion

We introduced RelabelMamba, an effective strategy for managing LNL across diverse datasets. The efficacy of RelabelMamba stems from its innovative integration of Mamba label rectification, AEKR strategy, and adaptive data augmentation techniques such as Puzzlemix and Adaptivemix. These elements collaborate to mitigate label noise, enhance the accuracy of the training data, and ensure robust feature extraction in environments with high label noise. Our approach not only improves label accuracy but also enhances the overall robustness and adaptability of the learning model, making it particularly effective for applications where label integrity is compromised. Additionally, experimental results demonstrate that RelabelMamba maintain high accuracy and stability across various levels of noise, exhibiting less performance degradation compared to other SOTA models. Notably, it excels on the CIFAR-10/100 datasets under diverse noise conditions and demonstrates superior learning capabilities on the more complex ANIMAL-10N and Clothing1M datasets, which are characterized by real-world noise.

CRediT authorship contribution statement

Ningwei Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Weiqiang Jin:** Writing – review & editing, Visualization, Investigation, Formal analysis, Conceptualization. **Shirou Jing:** Writing – review & editing, Visualization, Software, Investigation. **Haixia Bi:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Guang Yang:** Writing – review & editing, Validation, Funding acquisition.

Funding

This work is supported by NSFC under Grant 42201394 and Grant 12326615, the National Key R and D Program of China under Grant 2022YFA1003800, the Major Key Project of Peng Cheng Laboratory under Grant PCL2024A06, and the Qinchuangyuan High-level Innovation and Entrepreneurial Talent Program under Grant no. QCYRCXM-2022-30. Guang Yang was supported in part by the ERC IMI (101005122), the H2020 (952172), the MRC, UK (MC/PC/21013), the Royal Society, UK (IEC\NSFC\211235), the NVIDIA Academic Hardware Grant Program, the SABER project supported by Boehringer Ingelheim Ltd, Wellcome Leap Dynamic Resilience, and the UKRI Future Leaders Fellowship (MR/V023799/1).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] U.A. Bhatti, M. Huang, H. Neira-Molina, S. Marjan, M. Baryalai, H. Tang, G. Wu, S.U. Bazai, MFFCG–Multi feature fusion for hyperspectral image classification using graph attention network, Expert Syst. Appl. 229 (2023) 120496.

[2] Z. Zhang, Y. Ding, X. Zhao, L. Siye, N. Yang, Y. Cai, Y. Zhan, Multireceptive field: An adaptive path aggregation graph neural framework for hyperspectral image classification, Expert Syst. Appl. 217 (2023) 119508.

[3] R. Kaur, S. Singh, A comprehensive review of object detection with deep learning, Digit. Signal Process. 132 (2023) 103812.

[4] S.H. Yelleni, D. Kumari, P. Srijith, et al., Monte Carlo DropBlock for modeling uncertainty in object detection, Pattern Recognit. 146 (2024) 110003.

[5] J. Wu, W. Ji, H. Fu, M. Xu, Y. Jin, Y. Xu, MedSegDiff-V2: Diffusion-based medical image segmentation with transformer, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 6030–6038.

[6] J. Jain, J. Li, M.T. Chiu, A. Hassani, N. Orlov, H. Shi, Oneformer: One transformer to rule universal image segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 2989–2998.

[7] H.-Y. Zhou, J. Guo, Y. Zhang, X. Han, L. Yu, L. Wang, Y. Yu, Nnformer: Volumetric medical image segmentation via a 3d transformer, IEEE Trans. Image Process. (2023).

[8] S. Liu, S. Huang, S. Wang, K. Muhammad, P. Bellavista, J. Del Ser, Visual tracking in complex scenes: A location fusion mechanism based on the combination of multiple visual cognition flows, Inf. Fusion 96 (2023) 281–296.

[9] S.M. Marvasti-Zadeh, L. Cheng, H. Ghanei-Yakhdan, S. Kasaei, Deep learning for visual tracking: A comprehensive survey, IEEE Trans. Intell. Transp. Syst. 23 (5) (2021) 3943–3968.

[10] L. Jiang, D. Huang, M. Liu, W. Yang, Beyond synthetic noise: Deep learning on controlled noisy labels, in: International Conference on Machine Learning, PMLR, 2020, pp. 4804–4815.

[11] A. Gu, K. Goel, C. Ré, Efficiently modeling long sequences with structured state spaces, 2021, arXiv preprint arXiv:2111.00396.

[12] J.T. Smith, A. Warrington, S.W. Linderman, Simplified state space layers for sequence modeling, 2022, arXiv preprint arXiv:2208.04933.

[13] D.Y. Fu, T. Dao, K.K. Saab, A.W. Thomas, A. Rudra, C. Ré, Hungry hungry hippos: Towards language modeling with state space models, 2022, arXiv preprint arXiv:2212.14052.

[14] H. Mehta, A. Gupta, A. Cutkosky, B. Neyshabur, Long range language modeling via gated state spaces, 2022, arXiv preprint arXiv:2206.13947.

[15] Y. Xu, Z. Li, W. Li, Q. Du, C. Liu, Z. Fang, L. Zhai, Dual-channel residual network for hyperspectral image classification with noisy labels, IEEE Trans. Geosci. Remote Sens. 60 (2021) 1–11.

[16] T. Xiao, T. Xia, Y. Yang, C. Huang, X. Wang, Learning from massive noisy labeled data for image classification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 2691–2699.

[17] A. Ghosh, H. Kumar, P.S. Sastry, Robust loss functions under label noise for deep neural networks, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2017.

[18] Z. Zhang, M. Sabuncu, Generalized cross entropy loss for training deep neural networks with noisy labels, Adv. Neural Inf. Process. Syst. 31 (2018).

[19] Y. Kim, J. Yim, J. Yun, J. Kim, Nlnl: Negative learning for noisy labels, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 101–110.

[20] L. Jiang, Z. Zhou, T. Leung, L.-J. Li, L. Fei-Fei, Mentornet: Learning data-driven curriculum for very deep neural networks on corrupted labels, in: International Conference on Machine Learning, PMLR, 2018, pp. 2304–2313.

[21] B. Han, Q. Yao, X. Yu, G. Niu, M. Xu, W. Hu, I. Tsang, M. Sugiyama, Co-teaching: Robust training of deep neural networks with extremely noisy labels, Adv. Neural Inf. Process. Syst. 31 (2018).

[22] J. Li, R. Socher, S.C. Hoi, Dividemix: Learning with noisy labels as semi-supervised learning, 2020, arXiv preprint arXiv:2002.07394.

[23] H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, Mixup: Beyond empirical risk minimization, 2017, arXiv preprint arXiv:1710.09412.

[24] Z. Mai, G. Hu, D. Chen, F. Shen, H.T. Shen, Metamixup: Learning adaptive interpolation policy of mixup with metalearning, IEEE Trans. Neural Netw. Learn. Syst. 33 (7) (2021) 3050–3064.

[25] T. Pang, K. Xu, J. Zhu, Mixup inference: Better exploiting mixup to defend adversarial attacks, 2019, arXiv preprint arXiv:1909.11515.

[26] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.

[27] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114.

[28] D. Sinha, M. El-Sharkawy, Thin mobilenet: An enhanced mobilenet architecture, in: 2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference, UEMCON, IEEE, 2019, pp. 0280–0285.

[29] W. Jin, B. Zhao, Y. Zhang, J. Huang, H. Yu, WordTransABSA: Enhancing aspect-based sentiment analysis with masked language modeling for affective token prediction, Expert Syst. Appl. 238 (2024) 122289, http://dx.doi.org/10.1016/j.eswa.2023.122289.

[30] C. Szegedy, S. Ioffe, V. Vanhoucke, A. Alemi, Inception-v4, inception-resnet and the impact of residual connections on learning, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2017.

[31] D. Zhou, B. Kang, X. Jin, L. Yang, X. Lian, Z. Jiang, Q. Hou, J. Feng, Deepvit: Towards deeper vision transformer, 2021, arXiv preprint arXiv:2103.11886.

[32] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.

[33] C.-F.R. Chen, Q. Fan, R. Panda, Crossvit: Cross-attention multi-scale vision transformer for image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 357–366.

[34] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16000–16009.

[35] B. Zhao, W. Jin, Y. Zhang, S. Huang, G. Yang, Prompt learning for metonymy resolution: Enhancing performance with internal prior knowledge of pre-trained language models, Knowl.-Based Syst. 279 (2023) 110928, http://dx.doi.org/10.1016/j.knosys.2023.110928.

[36] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, 2023, arXiv preprint arXiv:2312.00752.

[37] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, Z. Shi, Rsmamba: Remote sensing image classification with state space model, 2024, arXiv preprint arXiv:2403.19654.

[38] H. Chen, J. Song, C. Han, J. Xia, N. Yokoya, Changemamba: Remote sensing change detection with spatio-temporal state space model, 2024, arXiv preprint arXiv:2404.03425.

[39] J. Ruan, S. Xiang, Vm-unet: Vision mamba unet for medical image segmentation, 2024, arXiv preprint arXiv:2402.02491.

[40] Z. Wang, C. Ma, Weak-Mamba-UNet: Visual mamba makes CNN and ViT work better for scribble-based medical image segmentation, 2024, arXiv preprint arXiv:2402.10887.

[41] Z. Xing, T. Ye, Y. Yang, G. Liu, L. Zhu, Segmamba: Long-range sequential modeling mamba for 3d medical image segmentation, 2024, arXiv preprint arXiv:2401.13560.

[42] H. Guo, J. Li, T. Dai, Z. Ouyang, X. Ren, S.-T. Xia, MambaIR: A simple baseline for image restoration with state-space model, 2024, arXiv preprint arXiv:2402.15648.

[43] J. Liu, R. Yu, Y. Wang, Y. Zheng, T. Deng, W. Ye, H. Wang, Point mamba: A novel point cloud backbone based on state space model with octree-based ordering strategy, 2024, arXiv preprint arXiv:2403.06467.

[44] T. Zhang, X. Li, H. Yuan, S. Ji, S. Yan, Point could Mamba: Point cloud learning via state space model, 2024, arXiv preprint arXiv:2403.00762.

[45] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, Y. Liu, Vmamba: Visual state space model, 2024, arXiv preprint arXiv:2401.10166.

[46] T. Guo, Y. Wang, C. Meng, Mambamorph: a mamba-based backbone with contrastive feature learning for deformable mr-ct registration, 2024, arXiv preprint arXiv:2401.13934.

[47] J. Ma, F. Li, B. Wang, U-mamba: Enhancing long-range dependency for biomedical image segmentation, 2024, arXiv preprint arXiv:2401.04722.

[48] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, X. Wang, Vision mamba: Efficient visual representation learning with bidirectional state space model, 2024, arXiv preprint arXiv:2401.09417.

[49] W. Zhang, Y. Wang, Y. Qiao, Metacleaner: Learning to hallucinate clean representations for noisy-labeled visual recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 7373–7382.

[50] K. Yi, J. Wu, Probabilistic end-to-end noise correction for learning with noisy labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 7017–7025.

[51] J. Goldberger, E. Ben-Reuven, Training deep neural-networks using a noise adaptation layer, in: International Conference on Learning Representations, 2016.

[52] Y. Wang, X. Ma, Z. Chen, Y. Luo, J. Yi, J. Bailey, Symmetric cross entropy for robust learning with noisy labels, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 322–330.

[53] J. Han, P. Luo, X. Wang, Deep self-learning from noisy labels, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 5138–5147.

[54] D. Arpit, S. Jastrzebski, N. Ballas, D. Krueger, E. Bengio, M.S. Kanwal, T. Maharaj, A. Fischer, A. Courville, Y. Bengio, et al., A closer look at memorization in deep networks, in: International Conference on Machine Learning, PMLR, 2017, pp. 233–242.

[55] X. Yu, B. Han, J. Yao, G. Niu, I. Tsang, M. Sugiyama, How does disagreement help generalization against label corruption? in: International Conference on Machine Learning, PMLR, 2019, pp. 7164–7173.

[56] S. Yun, D. Han, S.J. Oh, S. Chun, J. Choe, Y. Yoo, Cutmix: Regularization strategy to train strong classifiers with localizable features, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 6023–6032.

[57] J.-H. Kim, W. Choo, H.O. Song, Puzzle mix: Exploiting saliency and local statistics for optimal mixup, in: International Conference on Machine Learning, PMLR, 2020, pp. 5275–5285.

[58] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, C.A. Raffel, Mixmatch: A holistic approach to semi-supervised learning, Adv. Neural Inf. Process. Syst. 32 (2019).

[59] D. Hendrycks, N. Mu, E.D. Cubuk, B. Zoph, J. Gilmer, B. Lakshminarayanan, Augmix: A simple data processing method to improve robustness and uncertainty, 2019, arXiv preprint arXiv:1912.02781.

[60] Z. Liu, S. Li, D. Wu, Z. Liu, Z. Chen, L. Wu, S.Z. Li, Automix: Unveiling the power of mixup for stronger classifiers, in: European Conference on Computer Vision, Springer, 2022, pp. 441–458.

[61] Z. Huang, X. Bao, N. Zhang, Q. Zhang, X. Tu, B. Wu, X. Yang, IPMix: Label-preserving data augmentation method for training robust classifiers, Adv. Neural Inf. Process. Syst. 36 (2024).

[62] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.

[63] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, H. Jégou, Training data-efficient image transformers & distillation through attention, in: International Conference on Machine Learning, PMLR, 2021, pp. 10347–10357.

[64] J. Ni, D. Xiang, Z. Lin, C. López-Martínez, W. Hu, F. Zhang, DNN-based PolSAR image classification on noisy labels, IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 15 (2022) 3697–3713.

[65] A. Krizhevsky, G. Hinton, et al., Learning Multiple Layers of Features from Tiny Images, Technical Report TR-2009, University of Toronto, Toronto, ON, Canada, 2009, URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

[66] H. Song, M. Kim, J.-G. Lee, Selfie: Refurbishing unclean samples for robust deep learning, in: International Conference on Machine Learning, PMLR, 2019, pp. 5907–5915.

[67] Y. Xu, P. Cao, Y. Kong, Y. Wang, L_dmi: An information-theoretic noise-robust loss function, 2019, arXiv preprint arXiv:1909.03388.

[68] Q. Xia, F. Lee, Q. Chen, TCC-net: A two-stage training method with contradictory loss and co-teaching based on meta-learning for learning with noisy labels, Inform. Sci. 639 (2023) 119008.

[69] H. Song, N. Mitsuo, S. Uchida, D. Suehiro, No regret sample selection with noisy labels, Mach. Learn. (2024) 1–26.

[70] Y. Kim, J. Yun, H. Shon, J. Kim, Joint negative and positive learning for noisy labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 9442–9451.

[71] S. Zhang, Y. Li, Z. Wang, J. Li, C. Liu, Learning with noisy labels using hyperspherical margin weighting, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 16848–16856.

[72] D. Ortego, E. Arazo, P. Albert, N.E. O'Connor, K. McGuinness, Multi-objective interpolation training for robustness to label noise, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 6606–6615.

[73] J. Zhang, B. Song, H. Wang, B. Han, T. Liu, L. Liu, M. Sugiyama, BadLabel: A robust perspective on evaluating and enhancing label-noise learning, IEEE Trans. Pattern Anal. Mach. Intell. (2024).

[74] R. Higashimoto, S. Yoshida, M. Muneyasu, CRAS: Curriculum regularization and adaptive semi-supervised learning with noisy labels, Appl. Sci. 14 (3) (2024) 1208.

[75] J. Li, C. Xiong, S.C. Hoi, Learning from noisy data with robust representation learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 9485–9494.

[76] S. Liu, J. Niles-Weed, N. Razavian, C. Fernandez-Granda, Early-learning regularization prevents memorization of noisy labels, Adv. Neural Inf. Process. Syst. 33 (2020) 20331–20342.

[77] Z.-F. Wu, T. Wei, J. Jiang, C. Mao, M. Tang, Y.-F. Li, Ngc: A unified framework for learning with open-world noisy data, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 62–71.

[78] Z. Huang, J. Zhang, H. Shan, Twin contrastive learning with noisy labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 11661–11670.

[79] N. Karim, M.N. Rizve, N. Rahnavard, A. Mian, M. Shah, Unicon: Combating label noise through uniform selection and contrastive learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 9676–9686.

[80] C. Feng, Y. Ren, X. Xie, Ot-filter: An optimal transport filter for learning with noisy labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 16164–16174.

[81] F.R. Cordeiro, R. Sachdeva, V. Belagiannis, I. Reid, G. Carneiro, Longremix: Robust learning with high confidence samples in a noisy label environment, Pattern Recognit. 133 (2023) 109013.

[82] Q. Wei, L. Feng, H. Sun, R. Wang, C. Guo, Y. Yin, Fine-grained classification with noisy labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 11651–11660.

[83] Y. Tu, B. Zhang, Y. Li, L. Liu, J. Li, Y. Wang, C. Wang, C.R. Zhao, Learning from noisy labels with decoupled meta label purifier, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 19934–19943.

[84] K. Nishi, Y. Ding, A. Rich, T. Hollerer, Augmentation strategies for learning with noisy labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 8022–8031.

[85] H. Lin, Y. Li, Z. Zhang, L. Zhu, Y. Xu, Learning with noisy labels by semantic and feature space collaboration, IEEE Trans. Circuits Syst. Video Technol. (2024).

[86] E. Zheltonozhskii, C. Baskin, A. Mendelson, A.M. Bronstein, O. Litany, Contrast to divide: Self-supervised pre-training for learning with noisy labels, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 1657–1667.

[87] M. Chen, H. Cheng, Y. Du, M. Xu, W. Jiang, C. Wang, Two wrongs don't make a right: Combating confirmation bias in learning with label noise, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2023, pp. 14765–14773.

[88] F. Li, K. Li, J. Tian, J. Zhou, Regroup median loss for combating label noise, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 13474–13482.

[89] C.-C. Zong, Y.-W. Wang, M.-K. Xie, S.-J. Huang, Dirichlet-based prediction calibration for learning with noisy labels, 2024, arXiv preprint arXiv:2401.07062.

[90] A. Garg, C. Nguyen, R. Felix, T.-T. Do, G. Carneiro, Instance-dependent noisy label learning via graphical modelling, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2023, pp. 2288–2298.

[91] C. Feng, G. Tzimiropoulos, I. Patras, SSR: An efficient and robust framework for learning with unknown label noise, in: 33rd British Machine Vision Conference 2022, BMVC 2022, London, UK, November 21-24, 2022, BMVA Press, 2022, URL https://bmvc2022.mpi-inf.mpg.de/0372.pdf.

[92] H. Kang, S. Liu, H. Huang, J. Yu, B. Han, D. Wang, T. Liu, Unleashing the potential of regularization strategies in learning with noisy labels, 2023, arXiv preprint arXiv:2307.05025.

[93] Y. Chen, X. Shen, Y. Liu, Q. Tao, J.A. Suykens, Jigsaw-vit: Learning jigsaw puzzles in vision transformer, Pattern Recognit. Lett. 166 (2023) 53–60.

[94] Y. Zhang, S. Zheng, P. Wu, M. Goswami, C. Chen, Learning with feature-dependent label noise: A progressive approach, 2021, arXiv preprint arXiv:2103.07756.

[95] Y. Chen, X. Shen, S.X. Hu, J.A. Suykens, Boosting co-teaching with compression regularization for label noise, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 2688–2692.

[96] L. Deng, B. Yang, Z. Kang, J. Wu, S. Li, Y. Xiang, Separating hard clean samples from noisy samples with samples' learning risk for DNN when learning with noisy labels, Complex Intell. Syst. (2024) 1–22.

[97] S. Jiang, J. Li, J. Zhang, Y. Wang, T. Xu, Dynamic loss for robust learning, IEEE Trans. Pattern Anal. Mach. Intell. (2023).

[98] Y. Li, H. Han, S. Shan, X. Chen, Disc: Learning from noisy labels via dynamic instance-specific selection and correction, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 24070–24079.

[99] Y. Li, Y. Chen, X. Yu, D. Chen, X. Shen, SURE: Survey recipes for building reliable and robust deep networks, 2024, arXiv preprint arXiv:2403.00543.

[100] H. Kang, S. Liu, H. Huang, T. Liu, Channel-wise contrastive learning for learning with noisy labels, 2023, arXiv preprint arXiv:2308.06952.

[101] A. Garg, C. Nguyen, R. Felix, T.-T. Do, G. Carneiro, PASS: Peer-agreement based sample selection for training with noisy labels, 2023, arXiv preprint arXiv:2303.10802.

[102] J.-M. Guo, C.-C. Sun, K.-Y. Chan, C.-Y. Liu, Robust semi-supervised learning with multi-consistency and data augmentation, IEEE Trans. Consum. Electron. (2023).

[103] R. Sachdeva, F.R. Cordeiro, V. Belagiannis, I. Reid, G. Carneiro, ScanMix: learning from severe label noise via semantic clustering and semi-supervised learning, Pattern Recognit. 134 (2023) 109121.